

Exercise C1

Q1. If $\tan(\pi \cos \theta) = \cot(\pi \sin \theta)$, then the value(s) of $\cos\left(\theta - \frac{\pi}{4}\right)$ is (are)

- (A) $\frac{1}{2}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\pm \frac{1}{2\sqrt{2}}$ (D) None of these

Ans. C

Sol. -

Q2. The equation, $\sin^4 x - 2 \cos^2 x + a^2 = 0$ is solvable if :

- (A) $-\sqrt{3} \leq a \leq \sqrt{3}$ (B) $-\sqrt{2} \leq a \leq \sqrt{2}$ (C) $-1 \leq a \leq 1$ (D) None of these

Ans. B

Sol. -

Q3. The general solution of the equation, $\frac{1 - \sin x + \dots + (-1)^n \sin^n x + \dots}{1 + \sin x + \dots + \sin^n x + \dots} = \frac{1 - \cos 2x}{1 + \cos 2x}$ is :

- (A) $(-1)^n \left(\frac{\pi}{3}\right) + n\pi$ (B) $(-1)^n \left(\frac{\pi}{6}\right) + n\pi$ (C) $(-1)^{n+1} \left(\frac{\pi}{6}\right) + n\pi$
 (D) $(-1)^{n-1} \left(\frac{\pi}{3}\right) + n\pi, (n \in I)$

Ans. B

Sol. -

Q4. If the equation, $x^2 + 4 + 3 \sin(ax + b) - 2x = 0$ has atleast one real solution, where $a, b \in [0, 2\pi]$, then one possible value of $(a + b)$ can be equal to :

- (A) $\frac{7\pi}{2}$ (B) $\frac{5\pi}{2}$ (C) $\frac{9\pi}{2}$ (D) None of these

Ans. D

Sol. $x^2 + 4 + 3 \sin(ax + b) - 2x = 0$

$$\Rightarrow \frac{1}{3}[-x^2 + 2x - 4] = \sin(ax + b)$$

$$\therefore \sin(ax + b) \in [-1, 1]$$

$$\text{and } -\frac{x^2}{3} + \frac{2x}{3} - \frac{4}{3} = 0 \in \left(-\infty, -\frac{D}{4a}\right] \equiv (-\infty, -1]$$

Only possibility are $\underbrace{\sin(ax + b) = -1}_{\text{only for } x=1}$

$$\Rightarrow a + b = (4n - 1)\frac{\pi}{2}$$

$$\Rightarrow a + b = \frac{7\pi}{2} \quad \{a, b \in [0, 2\pi]\}$$

Q5. The number of points of intersection of the two curves , $y = 2 \sin x$ and $y = 5x^2 + 2x + 3$, is :

(A) Infinite (B) 0 (C) 1 (D) 2

Ans. B

Sol. -

Q6. The number of all possible triplets (a_1, a_2, a_3) such that $a_1 + a_2 \cos(2x) + a_3 \sin^2(x) = 0$ for all x is

(A) zero (B) one (C) three (D) infinite

Ans. D

Sol. -

Q7. The number of values of x in the interval , satisfying the equation $3 \sin^2 x - 7 \sin x + 2 = 0$ is

(A) 0 (B) 5 (C) 6 (D) 10

Ans. C

Sol. -

Q8. If $\sin^2 \theta - \sin \theta - 1 = 0$ is to be satisfied for exactly 4 distinct value of $\theta \in [0, n\pi]$, $n \in \mathbb{N}$, then least value of n is:

(A) 2 (B) 6 (C) 4 (D) 1

Ans. C

Sol. -

Q9. If $\tan(\cot x) = \cot(\tan x)$, then $\sin 2x =$

(A) $\frac{4}{(2n+1)\pi} n \in \mathbb{Z}$ (B) $\frac{4}{(2n+1)\pi} n \in \mathbb{Z} \setminus \{-1, 0\}$ (C) $\frac{4}{(2n+1)\pi} n \in \mathbb{Z} \setminus \{-1\}$

(D) $\frac{4}{(2n+1)\pi} n \in \mathbb{Z} - \{0\}$

Ans. B

Sol. -

Q10. The general solution of the equation $\tan^2 \alpha + 2\sqrt{3} \tan \alpha = 1$ is given by

(A) $\alpha = \frac{n\pi}{2}$ (B) $\alpha = (2n+1)\frac{\pi}{2}$ (C) $\alpha = (6n+1)\frac{\pi}{12}$ (D) $\alpha = n\frac{\pi}{12}$

Ans. C

Sol. -

Q11. Number of real values of x in $[-2\pi, 2\pi]$ for which value of the expression $y = \sqrt{2\sin x} - \sin x$ is maximum, is

(A) 0 (B) 2 (C) 4 (D) more than 4

Ans. C

Sol. -

Q12. The set of all x in $(-\pi, \pi)$ satisfying $|4\sin x - 1| < \sqrt{5}$ is given by

(A) $\left(-\frac{\pi}{10}, \frac{3\pi}{10}\right)$ (B) $\left(\frac{\pi}{10}, \frac{3\pi}{10}\right)$ (C) $\left(\frac{\pi}{10}, -\frac{3\pi}{10}\right)$ (D) None of these

Ans. A

Sol. -

Q13. The values of x between 0 and 2π , which satisfy the equation $\sin x \sqrt{8 \cos^2 x} = 0$ are in A.P. with common difference

(A) $\pi/4$ (B) $\pi/8$ (C) $3\pi/8$ (D) $5\pi/8$

Ans. A

Sol. -

Q14. α, β, γ and δ are the smallest positive angles in ascending order of magnitude which have their sines equal to the positive quantity k . The value of $4 \sin \frac{\alpha}{2} + 3 \sin \frac{\beta}{2} + 2 \sin \frac{\gamma}{2} + \sin \frac{\delta}{2}$ is equal to :

(A) $2\sqrt{1-5}$ (B) $2\sqrt{1+5}$ (C) $2\sqrt{5}$ (D) $2k$

Ans. B

Sol. -

Q15. The number of points inside or on the circle $x^2 + y^2 = 4$ satisfying $\tan^4 x + \cot^4 x + 1 = 3 \sin^2 y$ is

(A) one (B) two (C) four (D) infinite

Ans. C

Sol. -

Q16. $|\cos x| = \cos x - 2 \sin x$, if :

(A) $x = n\pi$ (B) $x = 2n\pi + \frac{\pi}{4}$ (C) $x = (2n+1)\pi$ (D) $x = (2n+1)\pi + \frac{\pi}{4}$

Ans. B, D

Sol. -

Q17. Consider the system of equations $\sin x \cos 2y = (a^2 - 1)^2 + 1$ and $\cos x \sin 2y = a + 1$.

Which of the following ordered pairs (x, y) of real numbers can satisfy the given system of equations for permissible real values of a ?

(A) $\left(\frac{-\pi}{2}, \frac{-\pi}{2}\right)$ (B) $\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$ (C) $\left(\frac{3\pi}{2}, \frac{-\pi}{2}\right)$ (D) $\left(\frac{-\pi}{2}, \frac{3\pi}{2}\right)$

Ans. A, C, D

Sol. -

Q18. If $(\cos^2 x + \sec^2 x)(1 + \tan^2 2y)(3 + \sin 3z) = 4$, then

(A) x may be a multiple of π (B) x cannot be an even multiple of π

(C) z can be a multiple of π (D) y can be a multiple of $\frac{\pi}{2}$

Ans. D, A

Sol. -

Q19. A solution (x, y) of the system of equations $x - y = 1/3$ and $\cos^2(\pi x) - \sin^2(\pi y) = 1/2$ is given by

(A) $(7/6, 5/6)$ (B) $(2/3, 1/3)$ (C) $(-5/6, -7/6)$ (D) $(13/6, 11/6)$

Ans. D, C, A

Sol. -

Q20. $\tan |x| = |\tan x|$ if :

(A) $x \in \left(-\frac{\pi(2k+1)}{2}, -\pi k\right]$ (B) $x \in \left[nk, \frac{\pi(2k+1)}{2}\right)$ (C) $x \in \left(-nk, -\frac{\pi(2k-1)}{2}\right)$

(D) $x \in \left(\frac{\pi(2k-1)}{2}, nk\right), k \in \mathbb{N}$

Ans. A, B

Sol. -